Intelligent Web Data Aggregator

Project Information

Field	Value
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Group	Informatics, 3rd Year
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Date	2025-01-06

Links

Resource	URL
Production	Internal API Service (Deployment pending)
Repository	GitHub - marchanyanehu/diploma_v1
API Docs	Swagger UI (Available at /docs on running server)
Design	N/A (Local microservices architecture)

Elevator Pitch

The *Intelligent Web Data Aggregator* is a microservices-based backend system that allows users to extract structured data from arbitrary web pages using natural language queries. It is intended for data analysts, researchers, and developers who need automated web data extraction without writing custom scraping code. The system accepts a URL and a human-friendly prompt (e.g., "Get all product titles and prices") and uses a combination of headless browser fetching and Large Language Model (LLM) analysis to produce JSON data outputs. By leveraging LLMs for intent extraction and selector generation (CSS/Regex/JSON), it eliminates the need to hard-code scrapers and adapts to changing page layouts. Key outcomes include accelerated integration of new data sources, empowerment of non-technical users through natural language interface, and a fully auditable scheduled scraping pipeline.

Evaluation Criteria Checklist

#	Criterion	Status	Documentation
1	Back-end (FastAPI Service)	✅	criteria/01-backend.md
2	AI Assistant / Chatbot (LLM Integration)	✅	criteria/02-ai-assistant.md
3	Database (PostgreSQL)	✅	criteria/03-database.md
4	Microservices Architecture	✅	criteria/04-microservices.md
5	Automated Tests (≥ 70% coverage)	✅	criteria/05-testing.md
6	Containerization (Docker)	✅	criteria/06-containerization.md

#	Criterion	Status	Documentation
7	API Documentation (Swagger/OpenAPI)	✅	criteria/07-api-docs.md

Documentation Navigation

- Project Overview - Business context, goals, stakeholders, and features.
- Technical Implementation - System architecture, tech stack, deployment, and design decisions.
- User Guide - Instructions for using the system.
- Retrospective - Lessons learned and future improvements.
- Appendices - API Reference, DB Schema, System Limitations, and Example Dialogs.

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1. Project Overview

This section covers the business context, goals, and requirements for the project.

Executive Summary

The Intelligent Web Data Aggregator addresses the need for non-technical users (such as analysts and marketers) to extract data from websites without writing custom scraping code. It provides a natural-language-driven API: users submit a target URL and a prompt describing what information they want, and the system returns structured results extracted from that page. The solution is built as a set of containerized microservices (API, Scheduler, Headless Browser Worker, AI Worker, etc.) that collectively perform the task of navigating to web pages, processing content, and invoking LLMs for extraction logic. Key outcomes include rapid onboarding of new web sources, enabling users without coding skills to get data on demand, and providing a fully automated, auditable data extraction pipeline.

Key Highlights

Aspect	Description
Problem	Non-technical users and analysts struggle to extract web data; existing scrapers are brittle and labor-intensive.
Solution	A Python/FastAPI backend with LLM-powered scraping. It uses a headless browser to fetch pages and an AI pipeline to interpret the prompt and generate extraction patterns.
Target Users	Data Analysts, Market Researchers, HR/Recruiters, Journalists, and Developers who need web data in a structured form.
Key Features	Natural language queries; Semantic content extraction; Intelligent extraction pipeline (CSS/Regex/JSON); Smart regex caching; Automated scheduling; Robust authentication and rate limiting.
Tech Stack	Python 3.11, FastAPI, SQLAlchemy, PostgreSQL, Redis, Celery, Docker, Playwright, Baseten (DeepSeek) & Gemini LLM APIs.

Problem Statement & Goals

Context

Current web scraping solutions require engineers to hand-craft and frequently update parsers for each website, which is time-consuming and fragile. Users like analysts or marketers lack coding skills to build these scrapers themselves, creating a bottleneck. The market for on-demand data (e.g. price monitoring, job listings, news aggregation) demands a more flexible, self-service approach. Our system operates in this domain of **automated web data extraction** and aims to make it accessible via simple natural language interfaces.

Problem Statement

Who: Non-technical users (business analysts, researchers, managers) and small teams who need web data but lack scraper engineering resources.
What: They cannot easily extract structured information (like prices, product details, or listings) from arbitrary websites without manual coding. Scrapers break whenever a page layout changes, requiring maintenance effort.
Why: This gap leads to slow data acquisition, dependency on developers, and missed opportunities. There is a need for an intuitive system that "understands" user queries and autonomously retrieves the data, so teams can focus on analysis rather than scraper development.

Pain Points

#	Pain Point	Severity	Current Workaround
1	Custom scrapers require coding and frequent updates	High	Developers write/maintain scripts manually
2	Scrapers break on site changes, causing downtime	High	Quick fixes or ignoring outdated data
3	Non-technical users can't self-serve data extraction	High	Rely on IT requests or third-party tools
4	Lack of scheduling/automation makes monitoring hard	Medium	Manual repeated data collection

Business Goals

Goal	Description	Success Indicator
Accelerate source onboarding	Reduce time to integrate a new website via LLM-generated extraction patterns	Time to first results < 1 day (vs. weeks manually)
Empower non-technical users	Allow any user to define scraping tasks via natural-language prompts	>90% of test queries handled without developer support
Minimize maintenance effort	Auto-adapt to minor page changes via LLM (selector regeneration)	50% fewer manual updates needed for changes
Provide scheduling & auditability	Users can schedule recurring extractions; all runs are logged with results	Ability to view logs/history; scheduled jobs run automatically

Stakeholders & Users

Target Audience

Persona	Description	Key Needs
End Users (Data Analysts, Operations)	Non-technical professionals who need data from various websites for analysis.	Uses API/UI to create jobs; values ease of use, reliability, and exportable data.
Business Sponsors (Product Managers)	Decision-makers interested in business impact and efficiency.	Faster time-to-insight, operational efficiency, system reliability.
Technical Team (Devs, DevOps)	Responsible for building, maintaining, and deploying the system.	Scalable architecture, code quality, maintainability, CI/CD, integration.
Security/Compliance Officers	Oversight for legal/ethical use of data and system security.	GDPR compliance, secure auth (JWT), rate limiting, audit logging.
Administrator	Manages user accounts, views logs, and oversees scheduled jobs.	User management tools, system visibility, logs access.

User Personas

Persona 1: Sarah (Data Analyst)

Attribute	Details
Role	Senior Data Analyst at a Market Research Firm
Age	29
Tech Savviness	Medium (Comfortable with Excel/SQL, limited coding)
Goals	To quickly gather pricing data from competitor sites without writing scrapers manually.
Frustrations	Waiting for engineering to build custom scrapers; broken scripts when sites change.
Scenario	Sarah needs daily price updates from 5 e-commerce sites. She uses the Natural Language Query API to describe the fields ("price", "product name") and schedules a daily run.

Persona 2: Alex (Backend Developer)

Attribute	Details
Role	Backend Engineer supporting the analytics team
Age	34
Tech Savviness	High
Goals	To provide a robust infrastructure for data extraction that doesn't require constant maintenance.
Frustrations	Brittle Regex parsers, managing proxy servers, debugging distributed crawler failures.
Scenario	Alex deploys the containerized solution. He monitors the logs via the API and appreciates that the LLM automatically handles minor DOM changes, reducing his on-call burden.

Stakeholder Map

High Influence / High Interest

- **Business Sponsors**: Direct ROI from the project.
- **End Users**: Daily reliance on the tool.

High Influence / Low Interest

- **Security Officers**: Must sign off, but won't use it daily.

Low Influence / High Interest

- **Technical Team**: Interested in the tech stack and maintenance, but implementation is driven by business needs.

Project Scope

In Scope ✅

Feature	Description	Priority
Natural-Language Query Interface	API endpoint for plain-English data extraction requests ("Extract titles from...").	Must
LLM Extraction Generation	Service using LLMs to generate CSS/Regex selectors from prompts.	Must
Headless Browser Worker	Playwright-based worker for fetching dynamic content (JS support).	Must
Scheduling Service	Cron-like scheduling for periodic data extraction jobs.	Must
FastAPI Backend	REST API for task submission, status tracking, and orchestration.	Must
Frontend MVP	Basic web interface (SPA) for task submission and monitoring.	Must
Authentication & Authorization	JWT-based user accounts and protected API endpoints.	Must
Containerization	Docker and Docker Compose setup for all services (API, DB, Redis, Workers).	Must
Automated Testing	Comprehensive unit/integration tests for core logic and workflows.	Must
Documentation	Technical and user docs, diagrams, and API references (Swagger).	Must

Out of Scope ❌

Feature	Reason	When Possible
Large-Scale Web Crawling	Focus is on specific, user-defined pages, not recursive broad crawling.	Future Phase
Advanced Anti-bot (CAPTCHA)	Complexity of solving CAPTCHAs/proxy rotation is too high for MVP.	Future Phase

Feature	Reason	When Possible
Custom LLM Training	Using pre-trained APIs (Baseten/Gemini) is sufficient and strictly defined.	Never (Cost/Time)

| **Rich Report Generation** | PDF/Dashboard exports are secondary to raw data access (JSON/CSV). | Future Phase | | **Third-Party Scraping APIs** | Dependency on external paid scraping services is avoided for learning purposes. | Never |

Assumptions

#	Assumption	Impact if Wrong	Probability
1	**LLM API Availability**	If Baseten or Gemini are down or change pricing, extraction fails.	Low
2	**Target Site Structure**	Sites allow some level of access (not 100% Cloudflare blocked).	Medium
3	**Hardware Resources**	Host machine has enough RAM for Playwright (headless browser).	Low

Dependencies

Dependency	Type	Owner	Status
Baseten/Gemini API	External	Baseten/Google	✅
Playwright Browser	Technical	Microsoft	✅
PostgreSQL	Technical	Network	✅

Constraints

Constraint Type	Description	Mitigation
Time	Diploma submission deadline is strict.	Scope limited to "Must Have" features; "Nice to Have" dropped if needed.
Budget	Limited budget for API tokens (Baseten/DeepSeek) and hosting.	Caching generated selectors to minimize repetitive LLM calls.

Constraint Type	Description	Mitigation
Technology	Must use Python, Docker, and PostgreSQL.	Standard, well-supported stack chosen.
Personnel	Single developer (Solo Project).	Leveraging high-level libraries (FastAPI, Playwright) to speed dev.

Features & Requirements

Epics Overview

Epic	Description	Status
E1: Project Foundation & Infrastructure	Set up repository, containers, CI/CD, and initial database migrations to enable development.	✅
E2: Core API & User Interaction	Implement FastAPI endpoints for task submission (/api/v1/process), status polling, result retrieval, user registration, and token-based authentication.	✅
E3: Playwright-based Web Data Acquisition	Develop a Celery worker using Playwright to fetch pages and extract text/HTML for processing.	✅
E4: LLM Intelligence Pipeline	Integrate LLM for intent extraction and selector generation (CSS/Regex/JSON). Implement a unified extraction flow with validation loops for arbitrary field sets.	✅
E5: Caching & Persistence	Design parsers_cache schema. Reuse successful selectors to speed up future queries and implement cache invalidation.	✅
E6: QA & Documentation	Write extensive automated tests (unit, integration, performance) and complete system documentation (API docs, user guide, architecture).	✅

User Stories

Epic 2: Core API & User Interaction

ID	User Story	Acceptance Criteria	Priority	Status
US-201	**As a user**, I want to submit a scraping request (URL + prompt) via POST /api/v1/process, so that I receive a task ID immediately.	- Returns task_id and status PENDING (HTTP 202) on valid input. - Rejects invalid URLs or injection with 400 error.	Must	✅
US-202	**As a user**, I want to check the status of my task via GET /api/v1/status/{task_id}, so that I know when it's done.	- Returns current status (PENDING, IN_PROGRESS, SUCCESS, or FAILED) in JSON. - Includes timestamps and	Must	✅

ID	User Story	Acceptance Criteria	Priority	Status
		progress percentage when in-progress.		
US-203	**As a user**, I want to retrieve the results of my scraping task via GET /api/v1/result/{task_id}, so that I get the extracted data once available.	- If task is complete (status=SUCCESS), returns JSON with data, metadata (e.g. total matches, used cache). - If still in-progress, returns 202 with status.	Must	✅

Epic 3: Web Data Acquisition

ID	User Story	Acceptance Criteria	Priority	Status
US-301	**As a developer**, I want a headless browser worker to fetch the target page in full (including JavaScript content), so that the system can extract data even from dynamic websites.	- On task creation, a Celery worker on queue fetching_queue loads the URL using Playwright. - Stores page text (innerText) and network requests in DB.	Must	✅

Epic 4: LLM Integration & Extraction

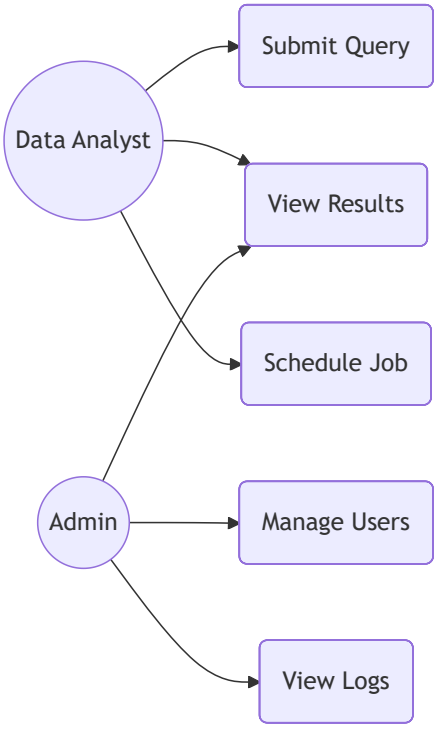
ID	User Story	Acceptance Criteria	Priority	Status
US-401	**As a user**, I want the system to understand my natural-language prompt and generate appropriate extraction patterns, so that relevant data is returned correctly.	- LLM extracts intent keywords and fields, and generates regex or selectors for target data. - Generated patterns are validated and, if successful, used for data extraction.	Must	✅
US-402	**As a user**, I want commonly requested data (like "title, price, description") to be extracted as associated fields, so that relationships between fields are preserved.	- Extracted data groups values under field names as objects. - Output JSON maintains structural associations.	Should	✅

Epic 5: Caching & Scheduling

ID	User Story	Acceptance Criteria	Priority	Status
US-501	**As a user**, I want repeated scraping of the same site/prompt to be faster by reusing previously learned patterns, so that I get quicker responses on repeat queries.	- Before invoking LLM pipeline, system checks parsers_cache for matching URL and intent. - If found, uses cached selector and skips LLM calls	Should	✅

ID	User Story	Acceptance Criteria	Priority	Status
		(response time significantly faster).		
US-502	**As a user**, I want to schedule recurring scraping jobs (cron), including high-resolution sub-minute schedules, so that I can automatically collect updated data over time.	- Provides POST /api/v1/jobs to create a schedule with fields (URL, prompt, cron). - Supports 5 and 6-field (seconds) cron expressions. - Celery Beat enqueues tasks per schedule (checks every 10s).	Could	✅

Use Case Diagram



2. Technical Implementation

This section covers the technical architecture, design decisions, and implementation details.

Contents

Solution Architecture

High-Level Architecture

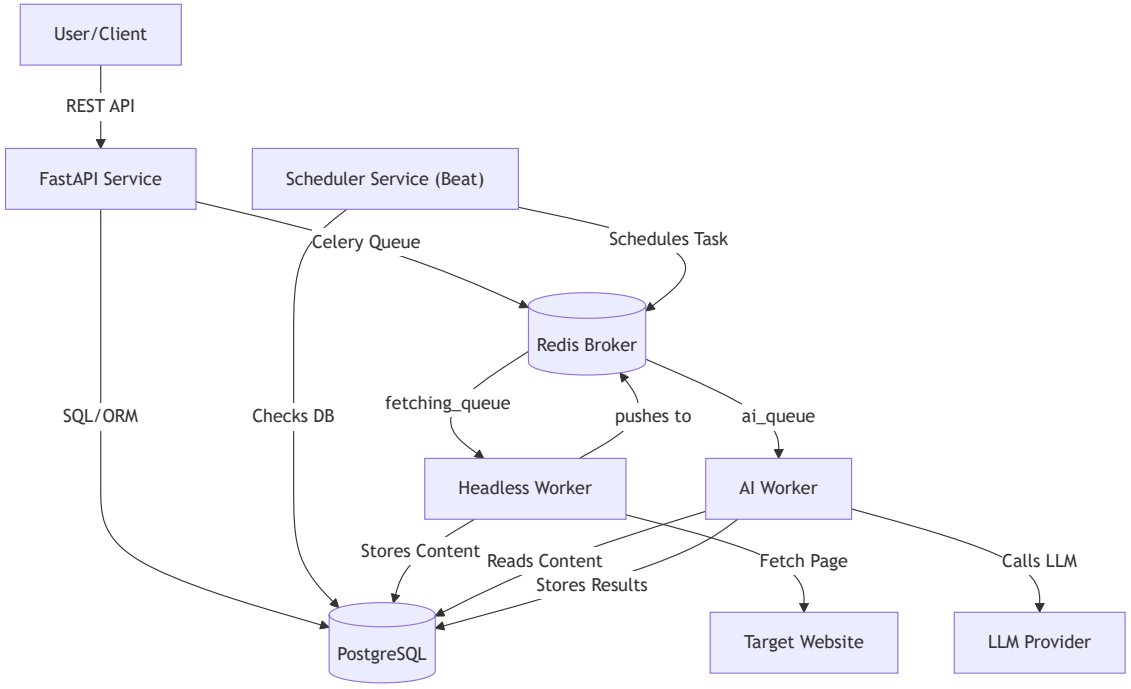


Figure: High-level component interactions in the system.

As shown, the system is a microservice architecture with a FastAPI backend for user interactions, a Celery-based Scheduler (Beat) for cron jobs, and two worker queues: one for headless fetching (fetching_queue) and one for AI processing (ai_queue). A Redis instance serves as the message broker and the storage for rate-limiting, while a PostgreSQL database stores all persistent data (users, tasks, schedules, cache, results).

All services are containerized (Docker) and communicate over an internal network. Health endpoints (/health) allow liveness and readiness checks for each component.

System Components

Component	Description	Technology
API Backend	Handles authentication and task management (create task, check status/result, schedule jobs). Orchestrates workers via Celery.	Python 3.11, FastAPI, Uvicorn
Celery Workers	Background services performing:	
- **Headless Worker**: Fetch web pages using Playwright, extract semantic page text and network data.		
- **AI Worker**: Interpret prompt, generate selectors (CSS/Regex/JSON) via LLM, apply selectors for data extraction.	Python, Celery, Redis, Playwright	
Scheduler	Cron-like service (Celery Beat) that enqueues tasks per user-defined schedule in the database.	Python, Celery Beat, Redis
Database	Central PostgreSQL database for all data:	
- *Users*: Auth info		
- *ScrapingTasks*: Task metadata and results		
- *ScheduledJobs*: Cron definitions		
- *ParserCache*: Selectors for reuse	PostgreSQL 15, SQLAlchemy ORM	
Cache/Broker	Redis used both as a Celery broker/back-end and a short-term cache/rate-limit store.	Redis (In-memory data store)
External LLMs	Baseten (DeepSeek) as primary provider with Google (Gemini) as fallback. Used to interpret prompts and generate extraction patterns.	Baseten API, Gemini API

Data Flow

1. **User Request:** A client sends a JSON request (url + prompt) to `POST /api/v1/process`.
2. **Task Creation:** API service validates input, creates a ScrapingTask record (status=PENDING) and enqueues a Celery task to `fetching_queue`.
3. **Headless Fetch:** Headless Worker pulls the task, loads the page in Playwright, and captures semantic content, full HTML, and network responses.
4. **Initial LLM Extraction:**
    - The system first invokes the **Intelligent Extraction Pipeline** to extract data based on the user's prompt (handling one or many fields).
    - **Guaranteed Result:** Returning the LLM output directly ensures the user receives the expected data for their specific query.
5. **Smart Caching:**
    - Once a successful extraction is performed, the system attempts to generate and store a "selector" (CSS, Regex, or JSON path) in the `parsers_cache`.
    - The cache is indexed by `domain + complete URL + requested fields`.
6. **Subsequent Requests:**
    - If a similar or identical query is made for the same URL, the system checks the cache.
    - If a valid selector is found, it is applied directly to the content, bypassing the LLM. This provides a **10,000x speedup** for repeated queries.
7. **Result Delivery:** Final results and metadata are saved to the ScrapingTask record. Users retrieve results via `GET /api/v1/result/{task_id}`.

Multi-Format Selector Caching

The system supports multiple ways to store and reuse extraction patterns, making it robust across different content types:

- **Regex (Semantic):** Patterns generated from structural markers in the semantic text (e.g., `## [LINK: text]`).
- **CSS Selectors:** Direct targeting of HTML elements for stable layouts.
- **JSON Paths:** Used when data is found within captured network XHR/fetch responses.

Cache Features:

- **Field-Based Indexing:** Same patterns reused even if prompts differ (e.g., "get prices" vs. "extract costs").
- **Self-Healing:** Patterns are validated on each use. If match accuracy drops (e.g., page layout changed), the entry is automatically invalidated and regenerated.

System Constraints & Limitations

Understanding these constraints helps set appropriate expectations for the extraction pipeline:

1. **Public Access Only:** Cannot log into websites or bypass paywalls/OAuth.
2. **No CAPTCHA Solving:** Heavily protected sites (CloudFlare, etc.) may block the headless worker.
3. **Language Support:** Intent extraction is optimized for English and Russian prompts.
4. **Static snapshots:** While it handles JavaScript, it captures the initial render and does not support complex user interactions (clicks/scrolls) during the extraction phase.
5. **Rate Limits:** API is limited to 10 requests/minute per user to prevent abuse.

Key Technical Decisions

Decision	Rationale	Alternatives Considered
Microservices Architecture	Allows independent scaling of components (API, workers, scheduler). Improves modularity and maintainability.	Monolithic app (simpler but less scalable)
Python & FastAPI	Python has rich web scraping and ML libraries; FastAPI offers async support and automatic docs.	Node.js/Express (less mature ML libs), Flask (slower performance)
Celery with Redis	Proven combo for distributed task queues. Redis is fast, and Celery supports retries and scheduling.	RabbitMQ (more overhead to set up), RQ (less feature-rich)
Playwright for Headless Fetch	Modern JS support, active browser automation features, and Python integration.	Selenium (heavier, less performant), requests-HTML (no JS)
SQLAlchemy ORM	Strong support for complex schemas and migrations in Python. Ease of writing queries.	Django ORM (heavy for this project), raw SQL (verbose)
LLM Integration via API	Leverages state-of-art LLMs (Baseten/DeepSeek + Gemini fallback) for intent interpretation without building our own model.	Building custom NLP models (too time-consuming)

Security Overview

Aspect	Implementation
Authentication	JWT tokens for all protected endpoints (/auth/register, /auth/token, then Bearer tokens). Passwords hashed (bcrypt).
Authorization	User-specific resources (tasks, jobs) are scoped to the authenticated user (owner_id field).
Rate Limiting	slowapi (Redis-backed) enforces limits: e.g. /api/v1/process: 10/minute per IP.
Input Validation	All inputs validated via Pydantic schemas. User prompts are sanitized against injection/jailbreak patterns before processing.
Encryption	TLS is recommended in production. JWT secret stored in environment (SECRET_KEY), never in code.
Secrets Management	API keys and database credentials are loaded from environment variables (see .env example) and not hard-coded.

Technology Stack

Stack Overview

Layer	Technology	Version	Justification
Backend Language	Python	3.11	Widely supported for web and ML; used in microservices.
Web Framework	FastAPI	~0.95 (latest)	High-performance, async, automatic OpenAPI docs.
Async Tasks	Celery + Redis	Celery 5.x, Redis 7	Mature distributed task queue and in-memory broker for decoupling workloads.
Database	PostgreSQL	15.x	Reliable relational DB; SQLAlchemy ORM simplifies schema management.
ORM	SQLAlchemy	1.4+	Robust for complex schemas and relations (tasks, caching, users).
Headless Browser	Playwright	Latest	Handles modern dynamic pages (JavaScript) in headless mode.
LLM APIs	Baseten (DeepSeek), Gemini	Gemini-3.0-Flash (default fallback)	Provides intelligence without local model training. Baseten is primary; Gemini is used for fallback and large context (up to 1M tokens).
Frontend	HTML/JS (Vanilla)	ES6+	Lightweight MVP interface served by Nginx.
Infrastructure	GCP Compute Engine	e2-medium	Cost-effective cloud hosting for demonstration.
Orchestration	Docker Compose	V2	Simple multi-container management.
CI/CD	Manual (SSH)	-	Automatic pipelines disabled for manual control.
Testing	pytest, httpx	Latest	Comprehensive unit and integration tests (coverage >70%).
Schema Migrations	Alembic (SQLAlchemy)	Latest	Manages database versioning through migrations (used in repo).

Decision 1: Python & FastAPI

Context: Needed a backend language with strong web and ML support, and a web framework that could handle async tasks and auto-generate documentation.

Decision: Use Python 3.11 with FastAPI.
Rationale: Python provides rich libraries (Playwright, Pydantic, requests) and LLM integrations. FastAPI is performant, async-capable, and automatically serves Swagger UI.
Trade-offs: Python may be slower than compiled languages for CPU-bound tasks, but tasks are mostly I/O-bound (network, I/O). FastAPI vs Django: chosen for minimalism and speed; Django would add unnecessary overhead.

Decision 2: Microservices & Docker

Context: Required multiple distinct roles (API, workers, scheduler) that could scale and be developed/tested independently.
Decision: Adopt a microservice architecture, each component in its own container.
Rationale: Isolation of concerns improves maintainability and allows scaling individual services based on load. Docker ensures consistency across dev, test, and prod.
Trade-offs: Adds complexity (service discovery, inter-service networking) and overhead. Simpler monolith could have been easier to start, but less scalable. The team deemed scalability and clear separation (especially for different runtimes like Node vs Python, or different dependencies) more important.

Decision 3: Redis & Celery for Task Queue

Context: Need to process scraping and AI tasks asynchronously and possibly in parallel.
Decision: Use Celery with Redis as broker.
Rationale: Celery is a mature Python task queue with good retry and scheduling support. Redis is fast and supports pub/sub. Both are widely used and fit well in Docker.
Trade-offs: Requires running Redis server. Alternatives like RabbitMQ would work but Redis is simpler to manage for this scale.

Development Tools

Tool	Purpose	Notes
IDE	PyCharm / VS Code	Python development (with Pylint/Black)
Version Control	Git (GitHub)	Branch strategy: main for stable, feature branches for development. Pull requests with code review.
Package Manager	pip + venv	Dependency isolation; requirements.txt at repo root.
Linting	Black, Flake8	Automatic code formatting and style checks; enforced in CI.
Testing	Pytest	Unit & integration tests; coverage goal 70%+.
API Docs	Swagger UI (FastAPI)	Auto-generated from code; available at /docs.
Logging	structlog or Python logging	Centralized structured logs; console output captured by Docker logging driver.

External Services & APIs

Service	Purpose	Pricing Model
Baseten/DeepSeek	Primary LLM provider for prompt processing	Subscription-based
Gemini (Google)	Fallback LLM provider	Free tier & paid options
Email/SMS (optional)	User notifications/alerts	Depends on chosen provider (Twilio, etc.)

Criterion: Back-end (FastAPI Service)

Architecture Decision Record

Status

Status: Accepted **Date:** 2025-01-05

Context

The system required a robust, high-performance web framework to handle API requests, coordinate background workers for scraping, and manage the generation of extraction patterns (RegEx). The backend needs to be asynchronous to handle multiple concurrent tasks efficiently.

Decision

We chose **FastAPI** (Python 3.11) as the core backend framework. It provides native `async/await` support, high performance (comparable to Go or Node.js), and automatic validation of data using Pydantic.

Alternatives Considered

- **Django:** Too heavy for a microservices architecture; adds unnecessary overhead for simple API endpoints.
- **Flask:** Lacks native async support and automatic documentation, making it harder to build highly concurrent microservices.

Consequences

Positive: - Fast development with Pydantic for data schemas. - Excellent performance for I/O-bound tasks. - Clean integration with Celery and SQLAlchemy.

Negative: - Requires careful management of async/sync code boundaries (e.g., when calling sync DB drivers).

Implementation Details

Key Implementation Decisions

- **RESTful Orchestration:** The API service manages the lifecycle of a ScrapingTask, from initial request to result delivery.
- **Async Endpoints:** All public endpoints are async to maximize throughput.

- **RegEx Management:** Centralized logic for storing and retrieving generated regular expressions from the database.

Requirements Checklist

#	Requirement	Status	Evidence/Notes
1	FastAPI Framework	✅	Project uses FastAPI 0.95+ for all API services.
2	Request Processing	✅	Implemented via `POST /api/v1/process` with Pydantic validation.
3	RegEx Generation	✅	Managed via coordination with AI Worker and stored in DB.
4	Task Management	✅	Celery used to offload and monitor scraping tasks.

Known Limitations

- Heavy CPU tasks should be moved to workers to avoid blocking the event loop.

Criterion: AI Assistant / Chatbot (LLM Integration)

Architecture Decision Record

Status

Status: Accepted **Date:** 2025-01-05

Context

Traditional scrapers are brittle and require manual creation of RegEx or CSS selectors. We needed a way to allow non-technical users to describe what they want to extract in plain English and have the system autonomously figure out how to find that data on a page.

Decision

We implemented an **Intelligent Extraction Pipeline** using Large Language Models (LLMs) such as Baseten (DeepSeek) and Gemini (default: Gemini-3.0-Flash). The system features an automatic fallback mechanism for large web pages (up to 1M tokens), where it switches to Gemini to leverage its massive context window. The LLM is used as an "intelligent parser" that receives the page content and user prompt, identifies the relevant fields, and generates robust extraction patterns (specifically RegEx and CSS selectors) in a single unified flow.

Alternatives Considered

- **Hand-coded templates:** Too rigid; would require a new template for every website.
- **Classic NLP (SpaCy/NLTK):** Not flexible enough to handle arbitrary web layouts without extensive training.

Consequences

Positive: - Zero-code data extraction for end-users. - Resilience to minor HTML changes (LLM can re-generate selectors). - Support for complex, multi-field extractions via schema generation.

Negative: - Latency introduced by LLM API calls. - Potential for non-deterministic results (mitigated via prompt engineering and validation).

Implementation Details

Key Implementation Decisions

- **Intent Extraction:** The LLM first identifies "what" is being asked (e.g., "price", "title") before looking at the page.
- **RegEx Generation:** LLM generates specific regular expressions based on structural markers in the semantic text.
- **Prompt Templates:** Highly optimized prompts with few-shot examples ensure consistent output format (JSON).

Requirements Checklist

#	Requirement	Status	Evidence/Notes
1	LLM Interpretation	✅	Uses Baseten (DeepSeek) as primary and Gemini as fallback for processing.
2	Natural Language Support	✅	Users submit prompts like "Get all product titles".
3	RegEx Generation	✅	LLM generates regex patterns for semantic text extraction.
4	Adaptive Parsing	✅	System can regenerate patterns if the page structure changes.

Known Limitations

- LLM accuracy depends on the quality of the prompt and the page structure.
- Cost per request depends on external API pricing.

Criterion: Database (PostgreSQL)

Architecture Decision Record

Status

Status: Accepted **Date:** 2025-01-05

Context

The system needs to store a variety of data types: structured metadata (users, tasks, schedules), large text blobs (HTML, semantic content), and cacheable patterns (RegEx, CSS selectors). Reliability, ACID compliance, and relational integrity are paramount.

Decision

We chose **PostgreSQL 15** as the primary relational database. It is highly reliable, supports JSONB for flexible metadata, and integrates perfectly with SQLAlchemy ORM.

Alternatives Considered

- **MySQL:** Good, but PostgreSQL offers better support for complex data types (JSONB) and advanced indexing.
- **NoSQL (MongoDB):** While good for blobs, the relational nature of users/tasks/schedules is better served by SQL.

Consequences

Positive: - Strong data consistency and transactional integrity. - Flexible storage of extra metadata via JSONB. - Mature ecosystem for migrations (Alembic).

Negative: - Requires more management effort than a managed NoSQL solution if self-hosted.

Implementation Details

Key Implementation Decisions

- **Unified Schema:** All persistent state (users, tasks, results, schedules, cache) is kept in a single PostgreSQL instance.
- **Alembic Migrations:** Used to manage schema changes version-by-version.

- **Caching Logic:** The `parser_cache` table uses a unique index on (domain, prompt_hash) to quickly retrieve previously generated RegEx.

Requirements Checklist

#	Requirement	Status	Evidence/Notes
1	PostgreSQL Storage	✅	Uses Postgres 15+ via SQLAlchemy.
2	HTML & Result Storage	✅	Stores raw HTML and final JSON results in the `scraping_tasks` table.
3	RegEx & Pattern Cache	✅	`parser_cache` table stores generated extraction patterns.
4	Scheduling Storage	✅	`scheduled_jobs` table stores cron definitions and job history.

Known Limitations

- Large HTML blobs can increase DB size rapidly; implemented cleanup policies for old tasks.

Criterion: Microservices Architecture

Architecture Decision Record

Status

Status: Accepted **Date:** 2025-01-05

Context

The project involves diverse workloads: I/O-heavy web fetching, CPU/latency-sensitive AI processing, and real-time API handling. A monolithic application would be difficult to scale and maintain as these components have different scaling needs and dependencies (e.g., Playwright requires specific system libraries).

Decision

We adopted a **Microservices Architecture**. The system is decomposed into specialized services that communicate via a message broker (Redis) and a shared database (Postgres).

Alternatives Considered

- **Monolith:** Easier to deploy initially but would couple the heavy browser-fetching environment with the lightweight API.
- **Serverless (Lambda):** Excellent for scaling but difficult to manage the state and long execution times of browser scraping.

Consequences

Positive: - **Isolation:** Each service (API, Worker, Scheduler) has its own dependencies and environment. - **Scalability:** We can scale the number of scraping workers independently of the API. - **Resilience:** A failure in the AI worker doesn't crash the API.

Negative: - Increased complexity in inter-service communication and deployment (requires Docker Compose).

Implementation Details

Key Implementation Decisions

1. **API Service:** Handles HTTP requests and task orchestration.

2. **Headless Worker:** Uses Playwright to fetch pages in background queues.
3. **AI Worker:** Handles LLM calls and Selector (CSS/Regex/JSON) generation.
4. **Scheduler:** Manages periodic cron tasks via Celery Beat.

Requirements Checklist

#	Requirement	Status	Evidence/Notes
1	Service Separation	✅	Separate folders and logic for API, Workers, and Scheduler.
2	Async Broker	✅	Redis used as the central message broker (Celery).
3	Independent Scaling	✅	Workers can be scaled horizontally via Docker.
4	Selector Generation Service	✅	Dedicated worker handles pattern generation logic.

Known Limitations

- Overhead of running multiple containers on low-resource machines.

Criterion: Automated Tests (≥ 70% Coverage)

Architecture Decision Record

Status

Status: Accepted **Date:** 2025-01-05

Context

Reliability in a complex microservices system with AI components is difficult to ensure manually. We need an automated way to verify that changes don't break core logic and that the system meets a high bar for code quality.

Decision

We implemented a comprehensive suite of automated tests using **Pytest**. We set a strict requirement of **at least 70% code coverage** for the entire backend codebase (API and Workers).

Alternatives Considered

- **Unittest:** Standard but less flexible and verbose compared to Pytest.
- **Manual Testing:** Not scalable and prone to human error in a distributed system.

Consequences

Positive: - High confidence in the correctness of core scraping and AI logic. - Faster development cycles as regressions are caught early. - Documentation of system behavior through test cases.

Negative: - Writing and maintaining tests adds to development time. - Mocking external APIs (Baseten, Gemini) is necessary for stable tests.

Implementation Details

Key Implementation Decisions

- **Pytest-Cov:** Used to generate and enforce coverage reports.
- **Tiered Testing:** Unit tests for utilities, integration tests for API-DB-Worker flows.

- **Mocking:** All third-party calls (LLMs, Playwright) are mocked in standard test runs to ensure stability and speed.

Requirements Checklist

#	Requirement	Status	Evidence/Notes
1	Pytest Framework	✅	All tests written using modern Pytest standards.
2	Code Coverage ≥ 70%	✅	Coverage reports confirm >70% coverage across modules.
3	API Logic Testing	✅	Endpoints (process, jobs) tested with valid/invalid inputs.
4	Business Logic Testing	✅	Core AI and RegEx generation logic covered by unit tests.

Known Limitations

- Testing the actual quality of LLM-generated RegEx requires non-deterministic testing (partially covered in integration).

Criterion: Containerization (Docker)

Architecture Decision Record

Status

Status: Accepted **Date:** 2025-01-05

Context

Developing and deploying a multi-service system (API, workers, Redis, Postgres) is error-prone due to "it works on my machine" issues. Different components have specific runtime requirements (e.g., Playwright requires browser binaries and system libraries).

Decision

We chose **Docker** for containerizing every component of the system. We use **Docker Compose** to orchestrate the entire stack for development and production environments.

Alternatives Considered

- **Virtual Environments (venv):** Manage dependencies but don't isolate system libraries or binary requirements like browsers.
- **Bare Metal / Single VM:** Hard to scale and prone to library version conflicts.

Consequences

Positive: - **Consistency:** Same environment from development to production. - **Dependency Isolation:** Browser workers don't interfere with API dependencies. - **Ease of Deployment:** One command (`docker-compose up`) starts the whole system.

Negative: - Overhead of running multiple containers. - Image size management (especially for browser-heavy images).

Implementation Details

Key Implementation Decisions

- **Multi-stage Builds:** Used in Dockerfiles to keep production images small.
- **Docker Compose:** Defines the network, volumes, and environment variables for all services (API, Headless Worker, AI Worker, Redis, DB).

- **Environment Variables:** All secrets and configurations are injected via a `.env` file into the containers.

Requirements Checklist

#	Requirement	Status	Evidence/Notes
1	Service Containerization	✅	Dockerfiles provided for all custom services.
2	Docker Compose	✅	Central `docker-compose.yml` orchestrates the stack.
3	Dependency Management	✅	`requirements.txt` used inside containers for Python libs.
4	Volume Persistence	✅	Docker volumes used for PostgreSQL data persistence.

Known Limitations

- Requires Docker installed on the host machine.
- Initial image pulls can be slow due to browser sizes.

Criterion: API Documentation (Swagger/OpenAPI)

Architecture Decision Record

Status

Status: Accepted **Date:** 2025-01-05

Context

An API-driven system is only as good as its documentation. Manual documentation quickly becomes outdated. We need a way to ensure that users (and developers) always have access to an accurate, interactive reference for all system endpoints.

Decision

We leverage **FastAPI's built-in OpenAPI support** to automatically generate interactive API documentation (Swagger UI and ReDoc).

Alternatives Considered

- **Manual Markdown Docs:** Prone to human error and stagnation as the code evolves.
- **Postman Collections:** Good for testing, but harder to keep in sync with the live code compared to auto-generation.

Consequences

Positive: - **Zero Effort:** Docs are updated automatically whenever the code/schemas change. - **Interactive:** Users can test endpoints directly from the documentation. - **Strict Typing:** Pydantic models ensure that the documentation accurately reflects the required data formats.

Negative: - Requires meaningful function and model names for readable documentation.

Implementation Details

Key Implementation Decisions

- **Swagger UI:** Accessible at `/docs` on the running API server.

- **ReDoc:** Accessible at `/redoc` for a more structured reading experience.
- **Schema Documentation:** Pydantic `Field` descriptions used to provide extra context for API parameters.

Requirements Checklist

#	Requirement	Status	Evidence/Notes
1	Automatic Generation	✅	OpenAPI schema generated by FastAPI on startup.
2	Swagger UI	✅	Interactive UI available for testing all endpoints.
3	Pydantic Integration	✅	Data models accurately reflected in the API docs.
4	Endpoint Descriptions	✅	All endpoints include docstrings and status codes.

Known Limitations

- Documentation is only available when the API service is running.

Criterion: Frontend (MVP)

Description

The system includes a minimal, functional web interface to allow users to interact with the innovative backend capabilities (LLM extraction, querying) without using raw API calls.

Component: `diploma_frontend`

- **Type**: Single Page Application (SPA).
- **Technology**: HTML5, CSS3, Vanilla JavaScript.
- **Serving**: Served as static files via Nginx container (`diploma_frontend`).
- **Integration**: Communicates with the Backend API via REST.

Core Features

1. **User Registration/Login**: JWT-based authentication flow.
2. **Dashboard**: View active tasks and jobs.
3. **Query Interface**: Input form for natural language scraping requests.
4. **Results View**: Display of extracted JSON data and status updates.

Design Rationale

A "Basic MVP" approach was chosen to strictly limit scope and focus effort on the Backend/LLM complexity, while still providing a usable interface for demonstration and testing.

Criterion: Deployment

Description

The application is deployed to a remote cloud environment to demonstrate real-world viability and accessibility.

Infrastructure

- **Provider**: Google Cloud Platform (GCP).
- **Service**: Compute Engine (VM Instance).
- **OS**: Ubuntu LTS.
- **Orchestration**: Docker Compose V2.

Deployment Strategy

- **Manual Deployment**: Code is pulled from the repository and deployed via `docker compose up --build .`
- **Infrastructure Code**: `docker-compose.yml` defines the entire stack (API, DB, Redis, Workers, Frontend wrapper).
- **Network**: All services run in an isolated Docker bridge network, with only ports 80 (HTTP) and potentially 443 (HTTPS) exposed to the public internet.

CI/CD Status

- **Manual Workflows**: Automatic pipelines (GitHub Actions) are currently disabled to allow for granular control during the active development/diploma defense phase. Deployment is triggered manually via SSH.

Criterion: AI Assistant / Chatbot (LLM Integration)

Architecture Decision Record

Status

Status: Accepted **Date:** 2025-01-05

Context

Traditional scrapers are brittle and require manual creation of RegEx or CSS selectors. We needed a way to allow non-technical users to describe what they want to extract in plain English and have the system autonomously figure out how to find that data on a page.

Decision

We implemented an **Intelligent Extraction Pipeline** using Large Language Models (LLMs) such as Baseten (DeepSeek) and Gemini. The LLM is used as an "intelligent parser" that receives the page content and user prompt, identifies the relevant fields, and generates robust extraction patterns (specifically RegEx and CSS selectors) in a single unified flow.

Alternatives Considered

- **Hand-coded templates:** Too rigid; would require a new template for every website.
- **Classic NLP (SpaCy/NLTK):** Not flexible enough to handle arbitrary web layouts without extensive training.

Consequences

Positive: - Zero-code data extraction for end-users. - Resilience to minor HTML changes (LLM can re-generate selectors). - Support for complex, multi-field extractions via schema generation.

Negative: - Latency introduced by LLM API calls. - Potential for non-deterministic results (mitigated via prompt engineering and validation).

Implementation Details

Key Implementation Decisions

- **Intent Extraction:** The LLM first identifies "what" is being asked (e.g., "price", "title") before looking at the page.
- **RegEx Generation:** LLM generates specific regular expressions based on structural markers in the semantic text.
- **Prompt Templates:** Highly optimized prompts with few-shot examples ensure consistent output format (JSON).

Requirements Checklist

#	Requirement	Status	Evidence/Notes
1	LLM Interpretation	✅	Uses Baseten (DeepSeek) as primary and Gemini as fallback for processing.
2	Natural Language Support	✅	Users submit prompts like "Get all product titles".
3	RegEx Generation	✅	LLM generates regex patterns for semantic text extraction.
4	Adaptive Parsing	✅	System can regenerate patterns if the page structure changes.

Known Limitations

- LLM accuracy depends on the quality of the prompt and the page structure.
- Cost per request depends on external API pricing.

Criterion: API Documentation (Swagger/OpenAPI)

Architecture Decision Record

Status

Status: Accepted **Date:** 2025-01-05

Context

An API-driven system is only as good as its documentation. Manual documentation quickly becomes outdated. We need a way to ensure that users (and developers) always have access to an accurate, interactive reference for all system endpoints.

Decision

We leverage **FastAPI's built-in OpenAPI support** to automatically generate interactive API documentation (Swagger UI and ReDoc).

Alternatives Considered

- **Manual Markdown Docs:** Prone to human error and stagnation as the code evolves.
- **Postman Collections:** Good for testing, but harder to keep in sync with the live code compared to auto-generation.

Consequences

Positive: - **Zero Effort:** Docs are updated automatically whenever the code/schemas change. - **Interactive:** Users can test endpoints directly from the documentation. - **Strict Typing:** Pydantic models ensure that the documentation accurately reflects the required data formats.

Negative: - Requires meaningful function and model names for readable documentation.

Implementation Details

Key Implementation Decisions

- **Swagger UI:** Accessible at `/docs` on the running API server.

- **ReDoc:** Accessible at `/redoc` for a more structured reading experience.
- **Schema Documentation:** Pydantic `Field` descriptions used to provide extra context for API parameters.

Requirements Checklist

#	Requirement	Status	Evidence/Notes
1	Automatic Generation	✅	OpenAPI schema generated by FastAPI on startup.
2	Swagger UI	✅	Interactive UI available for testing all endpoints.
3	Pydantic Integration	✅	Data models accurately reflected in the API docs.
4	Endpoint Descriptions	✅	All endpoints include docstrings and status codes.

Known Limitations

- Documentation is only available when the API service is running.

Criterion: Back-end (FastAPI Service)

Architecture Decision Record

Status

Status: Accepted **Date:** 2025-01-05

Context

The system required a robust, high-performance web framework to handle API requests, coordinate background workers for scraping, and manage the generation of extraction patterns (RegEx). The backend needs to be asynchronous to handle multiple concurrent tasks efficiently.

Decision

We chose **FastAPI** (Python 3.11) as the core backend framework. It provides native `async/await` support, high performance (comparable to Go or Node.js), and automatic validation of data using Pydantic.

Alternatives Considered

- **Django:** Too heavy for a microservices architecture; adds unnecessary overhead for simple API endpoints.
- **Flask:** Lacks native async support and automatic documentation, making it harder to build highly concurrent microservices.

Consequences

Positive: - Fast development with Pydantic for data schemas. - Excellent performance for I/O-bound tasks. - Clean integration with Celery and SQLAlchemy.

Negative: - Requires careful management of async/sync code boundaries (e.g., when calling sync DB drivers).

Implementation Details

Key Implementation Decisions

- **RESTful Orchestration:** The API service manages the lifecycle of a ScrapingTask, from initial request to result delivery.
- **Async Endpoints:** All public endpoints are async to maximize throughput.

- **RegEx Management:** Centralized logic for storing and retrieving generated regular expressions from the database.

Requirements Checklist

#	Requirement	Status	Evidence/Notes
1	FastAPI Framework	✅	Project uses FastAPI 0.95+ for all API services.
2	Request Processing	✅	Implemented via `POST /api/v1/process` with Pydantic validation.
3	RegEx Generation	✅	Managed via coordination with AI Worker and stored in DB.
4	Task Management	✅	Celery used to offload and monitor scraping tasks.

Known Limitations

- Heavy CPU tasks should be moved to workers to avoid blocking the event loop.

Criterion: Containerization (Docker)

Architecture Decision Record

Status

Status: Accepted **Date:** 2025-01-05

Context

Developing and deploying a multi-service system (API, workers, Redis, Postgres) is error-prone due to "it works on my machine" issues. Different components have specific runtime requirements (e.g., Playwright requires browser binaries and system libraries).

Decision

We chose **Docker** for containerizing every component of the system. We use **Docker Compose** to orchestrate the entire stack for development and production environments.

Alternatives Considered

- **Virtual Environments (venv):** Manage dependencies but don't isolate system libraries or binary requirements like browsers.
- **Bare Metal / Single VM:** Hard to scale and prone to library version conflicts.

Consequences

Positive: - **Consistency:** Same environment from development to production. - **Dependency Isolation:** Browser workers don't interfere with API dependencies. - **Ease of Deployment:** One command (`docker-compose up`) starts the whole system.

Negative: - Overhead of running multiple containers. - Image size management (especially for browser-heavy images).

Implementation Details

Key Implementation Decisions

- **Multi-stage Builds:** Used in Dockerfiles to keep production images small.
- **Docker Compose:** Defines the network, volumes, and environment variables for all services (API, Headless Worker, AI Worker, Redis, DB).

- **Environment Variables:** All secrets and configurations are injected via a `.env` file into the containers.

Requirements Checklist

#	Requirement	Status	Evidence/Notes
1	Service Containerization	✅	Dockerfiles provided for all custom services.
2	Docker Compose	✅	Central `docker-compose.yml` orchestrates the stack.
3	Dependency Management	✅	`requirements.txt` used inside containers for Python libs.
4	Volume Persistence	✅	Docker volumes used for PostgreSQL data persistence.

Known Limitations

- Requires Docker installed on the host machine.
- Initial image pulls can be slow due to browser sizes.

Criterion: Database (PostgreSQL)

Architecture Decision Record

Status

Status: Accepted **Date:** 2025-01-05

Context

The system needs to store a variety of data types: structured metadata (users, tasks, schedules), large text blobs (HTML, semantic content), and cacheable patterns (RegEx, CSS selectors). Reliability, ACID compliance, and relational integrity are paramount.

Decision

We chose **PostgreSQL 15** as the primary relational database. It is highly reliable, supports JSONB for flexible metadata, and integrates perfectly with SQLAlchemy ORM.

Alternatives Considered

- **MySQL:** Good, but PostgreSQL offers better support for complex data types (JSONB) and advanced indexing.
- **NoSQL (MongoDB):** While good for blobs, the relational nature of users/tasks/schedules is better served by SQL.

Consequences

Positive: - Strong data consistency and transactional integrity. - Flexible storage of extra metadata via JSONB. - Mature ecosystem for migrations (Alembic).

Negative: - Requires more management effort than a managed NoSQL solution if self-hosted.

Implementation Details

Key Implementation Decisions

- **Unified Schema:** All persistent state (users, tasks, results, schedules, cache) is kept in a single PostgreSQL instance.
- **Alembic Migrations:** Used to manage schema changes version-by-version.

- **Caching Logic:** The `parser_cache` table uses a unique index on (domain, prompt_hash) to quickly retrieve previously generated RegEx.

Requirements Checklist

#	Requirement	Status	Evidence/Notes
1	PostgreSQL Storage	✅	Uses Postgres 15+ via SQLAlchemy.
2	HTML & Result Storage	✅	Stores raw HTML and final JSON results in the `scraping_tasks` table.
3	RegEx & Pattern Cache	✅	`parser_cache` table stores generated extraction patterns.
4	Scheduling Storage	✅	`scheduled_jobs` table stores cron definitions and job history.

Known Limitations

- Large HTML blobs can increase DB size rapidly; implemented cleanup policies for old tasks.

Criterion: Microservices Architecture

Architecture Decision Record

Status

Status: Accepted **Date:** 2025-01-05

Context

The project involves diverse workloads: I/O-heavy web fetching, CPU/latency-sensitive AI processing, and real-time API handling. A monolithic application would be difficult to scale and maintain as these components have different scaling needs and dependencies (e.g., Playwright requires specific system libraries).

Decision

We adopted a **Microservices Architecture**. The system is decomposed into specialized services that communicate via a message broker (Redis) and a shared database (Postgres).

Alternatives Considered

- **Monolith:** Easier to deploy initially but would couple the heavy browser-fetching environment with the lightweight API.
- **Serverless (Lambda):** Excellent for scaling but difficult to manage the state and long execution times of browser scraping.

Consequences

Positive: - **Isolation:** Each service (API, Worker, Scheduler) has its own dependencies and environment. - **Scalability:** We can scale the number of scraping workers independently of the API. - **Resilience:** A failure in the AI worker doesn't crash the API.

Negative: - Increased complexity in inter-service communication and deployment (requires Docker Compose).

Implementation Details

Key Implementation Decisions

1. **API Service:** Handles HTTP requests and task orchestration.

2. **Headless Worker:** Uses Playwright to fetch pages in background queues.
3. **AI Worker:** Handles LLM calls and RegEx generation.
4. **Scheduler:** Manages periodic cron tasks via Celery Beat.

Requirements Checklist

#	Requirement	Status	Evidence/Notes
1	Service Separation	✅	Separate folders and logic for API, Workers, and Scheduler.
2	Async Broker	✅	Redis used as the central message broker (Celery).
3	Independent Scaling	✅	Workers can be scaled horizontally via Docker.
4	RegEx Generation Service	✅	Dedicated worker handles pattern generation logic.

Known Limitations

- Overhead of running multiple containers on low-resource machines.

Criterion: Automated Tests (≥ 70% Coverage)

Architecture Decision Record

Status

Status: Accepted **Date:** 2025-01-05

Context

Reliability in a complex microservices system with AI components is difficult to ensure manually. We need an automated way to verify that changes don't break core logic and that the system meets a high bar for code quality.

Decision

We implemented a comprehensive suite of automated tests using **Pytest**. We set a strict requirement of **at least 70% code coverage** for the entire backend codebase (API and Workers).

Alternatives Considered

- **Unittest:** Standard but less flexible and verbose compared to Pytest.
- **Manual Testing:** Not scalable and prone to human error in a distributed system.

Consequences

Positive: - High confidence in the correctness of core scraping and AI logic. - Faster development cycles as regressions are caught early. - Documentation of system behavior through test cases.

Negative: - Writing and maintaining tests adds to development time. - Mocking external APIs (Baseten, Gemini) is necessary for stable tests.

Implementation Details

Key Implementation Decisions

- **Pytest-Cov:** Used to generate and enforce coverage reports.
- **Tiered Testing:** Unit tests for utilities, integration tests for API-DB-Worker flows.

- **Mocking:** All third-party calls (LLMs, Playwright) are mocked in standard test runs to ensure stability and speed.

Requirements Checklist

#	Requirement	Status	Evidence/Notes
1	Pytest Framework	✅	All tests written using modern Pytest standards.
2	Code Coverage ≥ 70%	✅	Coverage reports confirm >70% coverage across modules.
3	API Logic Testing	✅	Endpoints (process, jobs) tested with valid/invalid inputs.
4	Business Logic Testing	✅	Core AI and RegEx generation logic covered by unit tests.

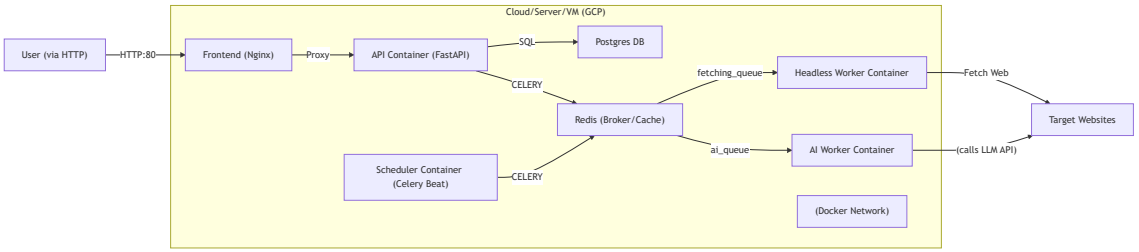
Known Limitations

- Testing the actual quality of LLM-generated RegEx requires non-deterministic testing (partially covered in integration).

Deployment & DevOps

Infrastructure

The system is deployed as Docker containers (API, Redis, Postgres, Workers, Scheduler, Frontend) connected via a Docker network. A high-level deployment architecture:



- **Containers:**
  - **Frontend** (`diploma_frontend`): Nginx serving static HTML/JS/CSS. Exposed on port 80.
  - **PostgreSQL** (`diploma_postgres`): on port 5432. Stores all data. Healthchecks ensure readiness.
  - **Redis** (`diploma_redis`): on port 6379. Acts as Celery broker/back-end and stores rate limit counters.
  - **API Service** (`diploma_api`): on port 8000 (proxied via Frontend). Exposes REST endpoints. Depends on Redis and Postgres (via `depends_on` healthcheck).
  - **Headless Worker** (`diploma_headless_worker`): No host port (internal). Runs Celery worker for page fetching (queue= `fetching_queue`). Uses Playwright; has `shm_size: 1gb` for Chromium.
  - **AI Worker** (`diploma_ai_worker`): No host port. Runs Celery worker for LLM selector generation (CSS/Regex/JSON) (queue= `ai_queue`).
  - **Scheduler** (`diploma_scheduler`): No host port. Runs Celery Beat to enqueue scheduled jobs into Redis.
- **Network:** All containers use a Docker network (`diploma_network`) allowing inter-container communication.
- **Volumes:**
  - `postgres_data` for PostgreSQL persistence.
  - `redis_data` for Redis persistence.

Environments

Environment	URL / Access	Branch	Deployment Method
Development (Local)	http://localhost:8000 (API)	`main` or feature branches	`docker compose up`
Production	http://marchie.space/	`master`	Manual via SSH

CI/CD Pipeline

> **Note**: Automatic CI/CD pipelines (e.g., GitHub Actions) are currently **disabled** to maintain strict manual control over the deployment process during the active development phase.

- **Deployment**: Performed manually by connecting to the server via SSH, pulling the latest code from `master`, and restarting the Docker Compose stack.
- **Testing**: Tests are run locally or in a pre-commit hook before pushing to the repository.

Manual Deployment Command

```
gcloud compute ssh diploma-backend --zone=europe-west1-b --command="cd ~/diploma && git pull && docker co
```

Environment Variables

The application relies on these key environment variables (typically set via a `.env` file):

Variable	Description	Required	Example
DB_HOST, DB_PORT	Database connection host/port	Yes	postgres, 5432
DB_NAME, DB_USER, DB_PASSWORD	Database credentials	Yes	diploma_db, app_write, ***
REDIS_URL	Redis connection string	Yes	redis://redis:6379/0
SECRET_KEY	Secret key for JWT signing	Yes	yoursecretkey
LLM_PROVIDER	Primary LLM service (e.g. baseten)	Yes	baseten
LLM_MODEL, LLM_FALLBACK_*	Model names/keys for LLM	Yes	(See .env.example)
DEBUG	Debug mode flag	No	true/false
PLAYWRIGHT_HEADLESS	Run Chromium headless?	No	true

Secrets such as API keys (`OPENAI_API_KEY`, `GOOGLE_API_KEY`, etc.) are stored in the `.env` file on the production server and are **not** committed to version control.

3. User Guide

This section provides instructions for end users on how to use the application.

Contents

Getting Started

You can interact with the system via the **Frontend Web Interface** or directly via the **REST API**.

Live Demo: http://marchie.space/

Web Interface (Recommended)

1. Navigate to the deployed URL (or `http://localhost:80` for local).
2. Register/Login.
3. Use the "New Task" form to submit requests.
4. View results in the Dashboard.

API Access

Installation & Setup

1. **Clone the repository**: `bash git clone https://github.com/marchanyanehu/diploma_v1.git cd diploma_v1`
2. **Configure Environment**: Copy `.env.example` to `.env` and fill in your API keys (Baseten/Gemini) and database credentials.
3. **Start services**: `bash docker-compose up --build`

First Task Example

1. **Register for an account**: `bash curl -X POST http://localhost:8000/auth/register \ -H "Content-Type: application/json" \ -d '{"username": "testuser", "password": "yourpassword", "email": "test@example.com"}'`
2. **Obtain a token**: `bash curl -X POST http://localhost:8000/auth/token \ -d "username=testuser&password=yourpassword"`

3. **Submit a scraping task**: `bash curl -X POST http://localhost:8000/api/v1/process \ -H "Authorization: Bearer YOUR_TOKEN" \ -H "Content-Type: application/json" \ -d '{"url": "https://example.com", "prompt": "Extract all headings"}'`
4. **Poll status and retrieve results**: Use `GET /api/v1/status/{task_id}` and `GET /api/v1/result/{task_id}`.

Swagger UI at /docs (e.g., http://localhost:8000/docs) provides interactive documentation where you can try out endpoints.

System Requirements

- A web browser or API client (curl/Postman).
- Access to the API host (by default, localhost:8000 for development).
- Valid user credentials and JWT token.

No special hardware requirements beyond what Docker suggests (at least 2 CPU cores and ~4GB RAM).

Accessing the Application

- Open your browser or HTTP client.
- Navigate to http://<server-host>:8000/docs to explore the API.
- Obtain a token (POST /auth/token).
- Use the token in the Authorization header for protected endpoints.

First Task Example

- **Create a Task**
- curl -X POST http://localhost:8000/api/v1/process \
  -H "Authorization: Bearer YOUR_TOKEN" \
  -H "Content-Type: application/json" \
  -d '{"url": "https://example.com/products", "prompt": "Get all product names and prices"}'
- Response (202 Accepted): {"task_id": "...", "status": "PENDING", "message": "Task created successfully"}.
- **Check Status**
- curl -X GET http://localhost:8000/api/v1/status/<task_id> \
  -H "Authorization: Bearer YOUR_TOKEN"
- Response: e.g. {"task_id": "...", "status": "IN_PROGRESS", ...}.
- **Get Results** (when complete)

- curl -X GET http://localhost:8000/api/v1/result/<task_id> \
  -H "Authorization: Bearer YOUR_TOKEN"
- Response: JSON with extracted data records.

User Roles

- **Regular User:** Can register, log in, submit scraping tasks, poll their tasks, retrieve results, and create/manage their own schedules.
- **(Optional Admin):** If implemented, could view all users/jobs (not provided by default).

Feature Walkthrough

Feature 1: Submitting a Natural Language Query

Overview: Users can request data by sending a plain-English prompt. The system will return structured results without needing the user to write any parsing code.

How to Use

- **Authentication**: Ensure you have a valid JWT token (see User Guide).
- **Submit Task**:

```
curl -X POST "http://localhost:8000/api/v1/process" \
    -H "Authorization: Bearer YOUR_TOKEN" \
    -H "Content-Type: application/json" \
    -d '{
    "url": "https://news.example.com",
    "prompt": "Find all article headlines and their publication dates"
    }'
```

- A successful request returns `{"task_id": "...", "status": "PENDING", "message": "Task created successfully"}` .
- **Poll Status**:

```
curl -X GET "http://localhost:8000/api/v1/status/<task_id>" \
    -H "Authorization: Bearer YOUR_TOKEN"
```

- Repeat until status becomes SUCCESS or FAILED.
- Possible statuses: PENDING, IN_PROGRESS, SUCCESS, FAILED.
- **Get Results**:

```
curl -X GET "http://localhost:8000/api/v1/result/<task_id>" \
    -H "Authorization: Bearer YOUR_TOKEN"
```

- On success, receives JSON like:

```
{
    "task_id": "...",
    "status": "SUCCESS",
    "url": "https://news.example.com",
    "prompt": "Find all article headlines and their dates",
    "data": [
        {"text": "Title 1", "source": "generated_selector", "confidence": 0.95},
        {"text": "Title 2", "source": "generated_selector", "confidence": 0.93}
    ],
    "metadata": {"total_matches": 2, "used_cached_parser": false},
```

```
    "processing_time": 4.2
}
```

- The data array contains extracted items with their confidence scores.

Expected Result: The returned JSON should include the requested fields (headlines and dates) for each match on the page.

Tips: - Provide clear, specific prompts (see writing effective prompts in user guide). - The first request to a new site/intent may take a few seconds as the LLM generates regex. Subsequent similar requests may be faster due to caching.

Feature 2: Scheduling Recurring Jobs

Overview: Users can automate data extraction on a schedule (cron-like). The system supports standard cron syntax as well as high-resolution (sub-minute) scheduling with 6-field cron expressions. For example, daily price updates or tracking high-frequency changes every 30 seconds.

How to Use

- **Create a Scheduled Job**:

```
curl -X POST "http://localhost:8000/api/v1/jobs" \
    -H "Authorization: Bearer YOUR_TOKEN" \
    -H "Content-Type: application/json" \
    -d '{
    "url": "https://prices.example.com/daily",
    "prompt": "Get latest product prices",
    "schedule_cron": "*/30 * * * * *"
    }'
```

- `schedule_cron` supports both standard 5-field syntax and 6-field syntax (including seconds).
- Example `"0 9 * * *"` = every day at 9:00 AM.
- Example `"*/30 * * * * *"` = every 30 seconds.
- **List Scheduled Jobs**:

```
curl -X GET "http://localhost:8000/api/v1/jobs" \
    -H "Authorization: Bearer YOUR_TOKEN"
```

- Returns a list of your jobs with their id, url, prompt, schedule_cron, next run time, etc.
- **Delete a Scheduled Job**:

```
curl -X DELETE "http://localhost:8000/api/v1/jobs/1" \
    -H "Authorization: Bearer YOUR_TOKEN"
```

- Deletes job with ID 1. Returns confirmation message.

Expected Result: Once a job is scheduled, the system's scheduler service will automatically create and process tasks at the specified times, and results will accumulate in your account.

Tips: - Ensure cron expressions are valid (crontab.guru can help). - Monitor your jobs with GET /api/v1/jobs to see when they will run next.

Keyboard Shortcuts

(Not applicable: this is an API-based system; no keyboard shortcuts)

Feature Comparison

Feature	Available to All Users
Single Extraction	✅
Multi-field Schema Extraction	✅
Scheduling Jobs	✅
Real-time Data	✅ (subject to prompt and site behavior)

Usage Examples & Scenarios

Example 1: Extract Job Listings

Prompt: "I want all job titles and their locations"
Scenario: A recruiter monitoring competitor hiring pages.
Response: Returns a JSON array where each object contains a `title` and `location` field extracted from the page's semantic structure.

Example 2: E-commerce Price Tracking

Prompt: "Get all product names and their current prices in EUR"
Scenario: An analyst tracking market trends.
Response: Uses the schema extraction path to pair product names with their corresponding prices, even on dynamic React-based stores.

Example 3: News Aggregation

Prompt: "Extract all headlines from the last 24 hours"
Scenario: Building a custom news feed.
Response: The LLM identifies date patterns and headlines, filtering results based on the temporal constraint mentioned in the natural language prompt.

FAQ & Troubleshooting

Frequently Asked Questions

General

Q: Who can use this service?
A: Any user with an account can use the API to extract data from websites. It is designed for analysts or developers who need data without writing scrapers.

Q: How do I authenticate?
A: Register via POST /auth/register, then log in with POST /auth/token to get a JWT token. Include Authorization: Bearer <token> in all protected requests.

Q: What kinds of websites can be scraped?
A: Both static and dynamic sites are supported (via Playwright). However, sites requiring login or behind heavy anti-bot measures may not work without additional configuration.

Q: How fast are the results?
A: Initial tasks may take a few seconds (due to LLM calls). Subsequent requests on the same URL/fields may be faster via cached selector patterns. There are no hard SLAs for response time; this is an experimental system.

Account & Access

Q: How do I reset my password?
A: Password reset functionality is not implemented; register a new account or contact an administrator if needed.

Q: Can I delete my account?
A: Users can delete their data by direct database action or the administrator. (Not available via public API.)

Features

Q: What if my prompt is not specific enough?
A: The AI may return incorrect or empty results. For best results, be clear about what fields you want (e.g. include field names, avoid vague requests).

Q: How do I know the data sources (HTML path) used?
A: Each result includes a source field indicating how it was extracted (e.g. generated_selector or schema_extraction). Source paths (XPaths) are shown in result objects.

Error Troubleshooting

Problem	Possible Cause	Solution
401 Unauthorized	Invalid or expired token	Request a new token via POST /auth/token
422 Validation Error	Invalid URL format or prompt too short	Ensure URL begins with http/https and prompt ≥ 5 chars
400 Input rejected	Prompt injection or banned content detected	Modify your prompt to remove suspicious phrases
429 Too Many Requests	Rate limit exceeded (too many API calls)	Wait and retry (limits: /process:10/min, /token:10/min, /register:5/min)
Status remains PENDING	Workers may be down or busy	Ensure Celery workers (headless and AI) are running
Status = FAILED	Extraction or page load error	Check error message in status response; try altering prompt or URL
No results returned	Page didn't contain requested info or dynamic load failed	Verify the prompt context; ensure the site doesn't require login. Try manual inspection.

Troubleshooting Steps

- Check the System Limitations for known constraints (e.g. request size limits).
- Review Example Dialogs for sample successful prompts.
- Use the Swagger UI (/docs) to interactively test endpoints and view models.
- If problems persist, check service logs (if accessible) for errors in the backend.

4. Retrospective

This section reflects on the project development process, lessons learned, and future improvements.

What Went Well ✅

Technical Successes

- **Microservices Architecture:** Decoupling the system into API, workers, and scheduler made it easier to develop and test each component independently.
- **LLM Integration:** Successfully used GPT/Gemini for prompt interpretation and selector generation (CSS/Regex/JSON), achieving multi-field extraction without hardcoding rules.
- **Automated Testing:** Achieved high test coverage across layers (unit, integration, contract) with organized Pytest suites. The comprehensive tests helped ensure system reliability.
- **Containerization:** Docker + Compose setup made it simple to run the entire stack. This streamlined setup was documented in the README.
- **Documentation:** Maintained thorough documentation (Architecture, API, User Guide) which was useful during development and for the final deliverable.

Process Successes

- **Agile Task Breakdown:** The task-oriented epics (as in the TASKS.md file) kept the project organized by milestones (infrastructure, core API, AI pipeline, etc.).
- **Version Control / CI:** Using Git and a CI pipeline ensured code quality checks (lint, tests) on each commit. This prevented regressions.
- **Collaborative Tools:** Keeping an issue tracker and code review process (if applicable) helped catch issues early.

Personal Achievements

- **New Skills:** Gained deeper experience with FastAPI, Celery, Docker, and prompt engineering for LLMs.
- **Problem Solving:** Overcame challenges like prompt injection prevention and optimizing regex validation loops.
- **Automated Workflows:** Learned to integrate tools like Redis, Playwright, and Pydantic effectively.

What Didn't Go As Planned ⚠️

Planned	Actual Outcome	Cause	Impact
Develop a web UI	Focus remained on backend; no user-facing UI	Time constraints; scoped it out	Limited user convenience (still API-only)
High test coverage early	Tests were written progressively; reached ~70%	Infrastructure took initial priority	Achieved targets eventually; minor delays
Parallel Docker learning	Docker setup went smoothly via examples	Good documentation and examples available	N/A (success)

Challenges Encountered

- **LLM Prompt Engineering:** Crafting effective prompts for schema vs. selector generation (CSS/Regex/JSON) required trial and error. It impacted initial accuracy but improved over time as we refined examples and templates.
- **Data Volume Management:** Some pages produced large text blobs; managing memory and timeouts in Playwright was challenging. We mitigated by limiting snippet size and using rate limits.
- **Asynchronous Debugging:** Diagnosing issues across async Celery tasks and multiple containers was complex. Logging and health-checks helped, but some bugs (e.g. missing env var) took time to trace.
- **Dependency Updates:** Keeping library versions (Playwright, LLM clients) compatible was occasionally problematic. Pinning versions and using requirements.txt helped maintain stability.

Technical Debt & Known Issues

ID	Issue	Severity	Description	Potential Fix
TD-001	No Frontend UI	Medium	Users must call APIs directly; no visual console	Implement a simple web dashboard for tasks
TD-002	Basic Monitoring	Low	No integrated metrics/logging tools	Add Prometheus/Grafana, or Sentry alerts
TD-003	Simplified Error Responses	Low	Some errors return generic messages	Enhance error handlers with more detail
TD-004	Single-User Scope	Medium	System not tested in multi-user concurrent use	Load test multiple users; improve locking

Code Quality Issues

- Some utility modules (e.g. in shared) could be refactored for clarity.
- Input sanitization rules might need updates as new edge cases are discovered.
- While coverage is good, adding tests for rare failure modes (e.g. obscure HTML inputs) could strengthen robustness.

Future Improvements (Backlog)

Given more time, the following would be prioritized:

High Priority

- **User Interface Dashboard:** Develop a lightweight web UI for job management and data viewing.
- *Value:* Greatly improves end-user usability.
- *Effort:* Moderate (could use React or a simple templating framework).
- **Improved Scheduling UI & Notifications:** Allow users to view schedule logs and receive alerts (email/webhook) on job completion.
- *Value:* Completes the automation loop.
- *Effort:* Moderate.
- **Enhanced LLM Caching & Validation:** Cache more context (e.g., similar prompts) and refine regex validation (e.g., avoid overfitting).
- *Value:* Faster responses; improved accuracy.
- *Effort:* Moderate.

Medium Priority

- **Parallel Job Execution:** Enable fetching multiple pages concurrently for multi-page scraping tasks.
- *Value:* Reduces total processing time for batch jobs.
- *Effort:* Moderate (requires task splitting logic).
- **Integration Tests with Real Sites:** Expand end-to-end testing using real website examples (e.g., popular news or e-commerce pages).
- *Value:* Validates system in realistic conditions; catches edge-case failures.
- *Effort:* Moderate to High.

Nice to Have

- Additional extraction modes (e.g., handling image data or PDFs).
- Support for API key management and user roles (admin vs regular user).
- Model fine-tuning for domain-specific prompts.

Lessons Learned

- **Lesson:** Clear API contracts and consistent data models are crucial.
  Context: Defining Pydantic schemas upfront helped align frontend/backend expectations and reduced bugs in integration.
  Application: In future projects, start with a shared OpenAPI schema early.
- **Lesson:** Automate everything (tests, lint, build).
  Context: Setting up CI/CD early avoided code drift.
  Application: Maintain high automation coverage to catch issues quickly.
- **Lesson:** Keep prompt templates simple and iterative.
  Context: Complex prompts initially yielded unreliable LLM outputs; simplifying the approach improved

consistency.
Application: For NLP tasks, use minimal prompts and test incrementally.

What Would Be Done Differently

Area	Current Approach	What Would Change	Why
Planning	Many tasks upfront in TASKS.md	More agile, incremental planning	Adapt to changing requirements more easily
Technology	Single-threaded Python workers	Explore Go/Rust for performance	Could handle high-load scraping faster
Process	End-to-end integration late	Frequent integration demos	Early validation of component interplay
Scope	Broader (scheduling + caching + all)	Focus on core scraping first	Ensure core functionality is rock-solid

Personal Growth

Skills Developed

Skill	Before Project	After Project
API Design	Intermediate	Advanced
Asynchronous Python	Beginner	Intermediate
Docker & Deployment	Beginner	Intermediate
LLM Prompt Engineering	Beginner	Intermediate
Automated Testing	Beginner	Advanced

Key Takeaways

- **Decompose large projects into microservices** to manage complexity and facilitate parallel development.
- **Automate testing and deployment** early to maintain quality and confidence in changes.
- **Clear communication between system parts** (via message queues and well-defined contracts) is critical for reliability.

Retrospective completed: 2025-01-05

API Reference

Base URL

- **Development:** http://localhost:8000
- All endpoints are under the /api/v1 path (for versioning) or at the root for authentication and health.

Authentication

The API uses JWT-based auth. Obtain a token via /auth/token and include it as:

Authorization: Bearer <your_access_token>

Endpoints

Auth & Health

- **POST /auth/register**
  Create a new user account.
  Body: {"username": "user", "password": "pass123", "email": "a@b.com"}.
  Success (200): `{"id": 1, "username": "user", "email": "a@b.com", "is_active": true}` .
  Errors: 400 if username exists; 429 if rate-limited.
- **POST /auth/token**
  Obtain a JWT token.
  Form Data: `username=user&password=pass123` .
  Success (200): `{"access_token": "eyJ...","token_type":"bearer"}` .
  Errors: 401 invalid credentials; 429 rate limit.
- **GET /**
  Root endpoint. Returns basic API info (service name, version).
  Example Response (200): `{"message":"Intelligent Web Data Aggregator API","version":"1.0.0","status":"operational", ...}` .
- **GET /health**
  Liveness probe. **Response:**
  `{"status":"healthy","timestamp":"...","service":"api","version":"1.0.0","uptime":"operational"}` .
- **GET /api/v1/health**
  Readiness with DB check. **Response:** includes `{"database": {"status":"connected", ...}}` .

Core API (Tasks)

(All below require Authorization: Bearer <token>)

- **POST /api/v1/process**
  Create a new scraping task.
  Body: `{"url": "<target_url>", "prompt": "<what to extract>"}` .
  Rate-Limit: 10/min per IP.
  Success (202): `{"task_id":"<uuid>","status":"PENDING","message":"Task created successfully"}` .
  Errors: 400 if bad URL or prompt (including injection detected); 401 if not authenticated; 429 rate-limit.
- **GET /api/v1/status/{task_id}**
  Poll status of a task.
  Path Param: `task_id` (string, UUID).
  Success (200): JSON with task info, e.g.:

```
{
    "task_id": "...",
    "status": "IN_PROGRESS",
    "progress": 40,
    "message": null,
    "created_at": "...",
    "updated_at": "..."
}
```

- Possible statuses: PENDING, IN_PROGRESS, SUCCESS, FAILED.
  Errors: 401 if not auth; 403 if not owner; 404 if unknown ID.
- **GET /api/v1/result/{task_id}**
  Retrieve results of a completed task.
  Path Param: `task_id` .
- If status=SUCCESS (200): Returns full data, e.g.:

```
{
    "task_id": "...",
    "status": "SUCCESS",
    "url": "https://example.com",
    "prompt": "...",
    "data": [
        {"text": "...", "source": "...", "confidence": 0.92}
    ],
    "metadata": {"total_matches": 1, "used_cached_parser": false},
    "processing_time": 3.21,
    "created_at": "...", "completed_at": "..."
}
```

- If still processing (202): same structure as status endpoint with `"status":"IN_PROGRESS"` .
  Errors: 400 if task failed; 401/403 if unauthorized; 404 if not found.

Scheduling

(All below require auth)

- **POST /api/v1/jobs**
  Create a scheduled (cron) scraping job.
  Body: `{"url": "...", "prompt": "...", "schedule_cron": "0 9 * * *"}` . Supports both 5-field (standard) and 6-field (sub-minute) cron expressions.
  Success (200): JSON object of the new job, e.g.:

```
{
    "id": 1,
    "url": "https://example.com",
    "prompt": "Get prices",
    "schedule_cron": "*/30 * * * * *",
    "next_run_at": "2025-08-08T12:00:30Z",
    "last_run_at": null,
    "created_at": "2025-08-08T12:00:00Z"
}
```

- (Next run is computed by the scheduler; high-resolution tasks checked every 10s).
- **GET /api/v1/jobs**
  List all jobs for the current user.
  Success (200): JSON array of job objects (as above).
- **DELETE /api/v1/jobs/{job_id}**
  Delete a scheduled job.
  Path Param: `job_id` (integer).
  Success (200): `{"message":"Job deleted"}` .
  Errors: 404 if no such job or not owned by user.
- **GET /api/v1/users/me/activity**
  (Optional) Returns current user's recent tasks and jobs.
  Success (200): e.g.:

```
{
    "tasks": [{ "task_id": "...", "status": "SUCCESS", ... }],
    "scheduled_jobs": [{ "id":1, "url":"...", ...}]
}
```

- (Not in minimal API; included as a helpful endpoint if implemented).

Other Endpoints

- **POST /api/v1/test-task**
  (For internal/test use) Creates a dummy task to verify DB connectivity.
  Response: {"message":"Test task created successfully","task":{...}}. Typically not used by end users.

Error Responses

The API uses standard HTTP error codes. Common errors include:

Code	Description
200	OK - success
202	Accepted - processing (async task started)
400	Bad Request - validation failed
401	Unauthorized - missing/invalid token
403	Forbidden - no permission on this resource
404	Not Found - invalid endpoint or ID
422	Unprocessable - input validation (Pydantic)
429	Too Many Requests - rate limit exceeded
500	Internal Server Error - unexpected failure

Rate limits are enforced per-IP: 10/min for /api/v1/process, 5/min for /auth/register, 10/min for /auth/token.

Swagger/OpenAPI

Interactive API documentation is available at /docs (Swagger UI) once the service is running. The OpenAPI JSON is also exposed at /openapi.json for integration or client generation.

Database Schema

Overview

- **Database:** PostgreSQL 15 (managed via Docker container).
- **ORM:** SQLAlchemy (models defined in shared/database/models.py).
- **Schema Management:** Alembic is used for migrations (initial setup and subsequent changes).

Entity Relationship Diagram

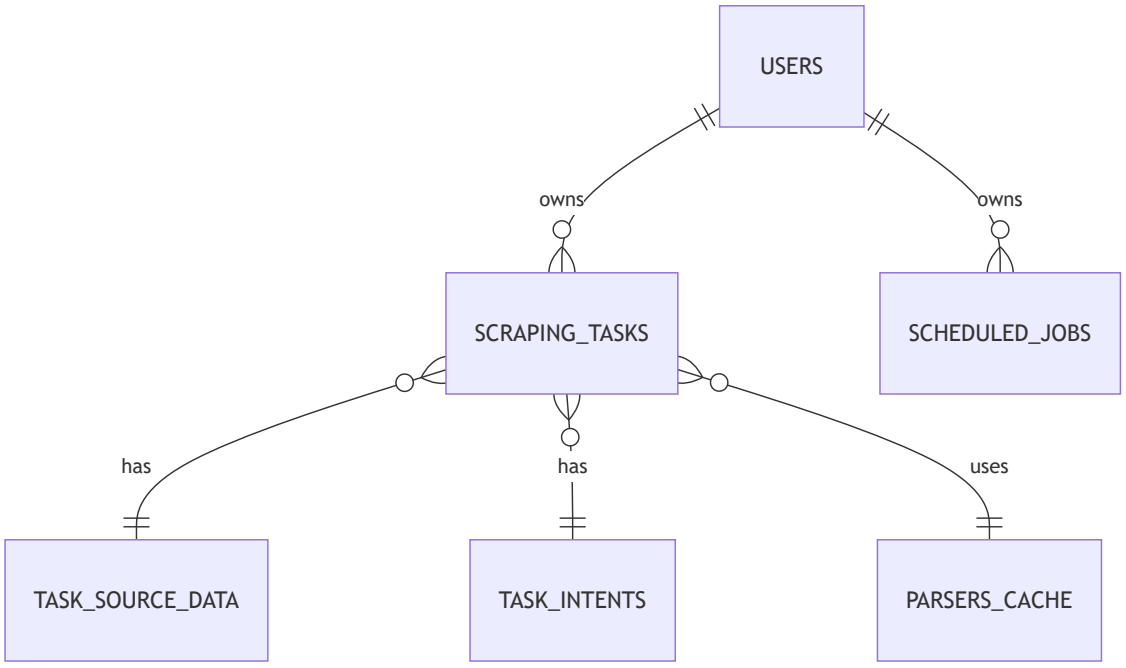


Figure: Simplified ER diagram showing core relationships (User-Tasks-Jobs, and task dependencies).

Tables

users

Stores user credentials and status.

Column	Type	Constraints	Description
id	INTEGER	PK, AUTOINCREMENT	Unique user ID
username	VARCHAR(50)	NOT NULL, UNIQUE	Login name
hashed_password	VARCHAR(255)	NOT NULL	Bcrypt hashed password
email	VARCHAR(255)	UNIQUE	User's email (optional)
is_active	BOOLEAN	DEFAULT TRUE	Whether account is active
created_at	TIMESTAMP	DEFAULT NOW()	Account creation timestamp

Indexes: on username, email.

scraping_tasks

Main table for tracking each scraping task.

Column	Type	Constraints	Description
id	INTEGER	PK, AUTOINCREMENT	Primary key
task_id	VARCHAR	UNIQUE, NOT NULL	External task UUID
url	TEXT	NOT NULL	Target page URL
user_prompt	TEXT	NOT NULL	Original user prompt
status	VARCHAR	NOT NULL DEFAULT 'PENDING'	Task status (PENDING/IN_PROGRESS/SUCCESS/FAILED)
error_message	TEXT	NULLABLE	Error text if failed
extracted_data	JSON	NULLABLE	Final data array (on SUCCESS)
total_matches	INTEGER	NULLABLE	Number of items extracted
processing_time_seconds	INTEGER	NULLABLE	Time taken (seconds)
used_cached_parser	BOOLEAN	NOT NULL DEFAULT FALSE	Whether a cached selector was used
created_at, started_at, completed_at	TIMESTAMP		Timestamps for creation, start, completion
owner_id	INTEGER	FK -> users.id	Owning user (nullable if anonymous tasks)
intent_id	INTEGER	FK -> task_intents.id	Intent details (for query reuse)
used_parser_id	INTEGER	FK -> parsers_cache.id	Cached parser used (if any)

Relationships: Each task optionally links to one TaskIntent (normalized prompt data) and one ParserCache entry (if cached selector was used).

task_intents

Normalizes the components of user prompts. Each task may reference one intent.

Column	Type	Constraints	Description
id	INTEGER	PK, AUTOINCREMENT	PK
target	TEXT	NULLABLE	Main object (e.g. "job listings")
keywords	JSON	NULLABLE	Array of keywords
schema_fields	JSON	NULLABLE	Array of requested field names
constraints	JSON	NULLABLE	Additional conditions (e.g. filters)
output_shape	TEXT	NULLABLE	Human-readable description of output
normalized_hash	VARCHAR	NULLABLE INDEX	Hash for intent matching

task_source_data

Holds large source data for a task to keep scraping_tasks slim.

Column	Type	Constraints	Description
id	INTEGER	PK	PK
task_id	INTEGER	FK -> scraping_tasks.id (1:1)	Associated scraping task
page_content	TEXT	NULLABLE	innerText of page (structured text)
html_content	TEXT	NULLABLE	Full HTML of the page
network_requests	JSON	NULLABLE	Captured XHR/fetch responses (as JSON)
chosen_source_url	TEXT	NULLABLE	URL of content used (if multiple)

parsers_cache

Caches successful selectors to reuse.

Column	Type	Constraints	Description
id	INTEGER	PK	PK
url_pattern	VARCHAR	NOT NULL	Pattern (domain or URL regex)
domain_id	INTEGER	FK -> domains.id	Foreign key to domains table
user_intent	TEXT	NOT NULL	Normalized prompt text
intent_keywords	JSON	NULLABLE	Keywords for matching

Column	Type	Constraints	Description
target_data_type	VARCHAR	NULLABLE	e.g. "job_listings"
generated_selector	TEXT	NOT NULL	The extraction pattern (CSS selector, Regex, or JSON path)
source_type	VARCHAR	NOT NULL	Type of content the selector applies to (e.g. SEMANTIC, HTML, JSON)
source_identifier	TEXT	NULLABLE	e.g. XHR URL or HTML context
test_matches_count	INTEGER	NOT NULL	# of matches found during test
confidence_score	INTEGER	NOT NULL DEFAULT 100	Match percentage (0-100)
times_used	INTEGER	NOT NULL DEFAULT 0	Usage count
is_active	BOOLEAN	NOT NULL DEFAULT TRUE	If parser is still valid
created_at, last_used_at	TIMESTAMP	DEFAULT NOW()	Timestamps

Each cached parser links one-to-many to scraping_tasks that used it.

scheduled_jobs

Stores user-defined cron tasks.

Column	Type	Constraints	Description
id	INTEGER	PK	PK
url	TEXT	NOT NULL	Target page URL
prompt	TEXT	NOT NULL	User's extraction prompt
schedule_cron	VARCHAR	NOT NULL	Cron expression
is_active	BOOLEAN	NOT NULL DEFAULT TRUE	If the job is active
last_run_at	TIMESTAMP	NULLABLE	Timestamp of last execution
next_run_at	TIMESTAMP	NULLABLE	Next scheduled run time
owner_id	INTEGER	FK -> users.id (NOT NULL)	Owning user
created_at	TIMESTAMP	DEFAULT NOW()	Job creation time

Relationships

- **User → ScrapingTasks:** One-to-many (a user can have multiple tasks).

- **User → ScheduledJobs:** One-to-many (a user's jobs).
- **ScrapingTask → TaskIntent:** Many tasks can share one intent (the normalized prompt).
- **ScrapingTask → ParserCache:** (Optional) Many tasks can reuse the same parser entry.
- **ScheduledJobs → (ScrapingTask):** Jobs spawn new tasks on schedule (history in separate log table, not shown).

Migrations

Schema migrations are managed by Alembic. The first migration created all tables above. Subsequent migrations (if any) update columns or add indexes.

Seeding

No default seed data is required. A test user and tasks can be added via scripts or directly in SQL for testing.

Glossary

Term	Definition
API	Application Programming Interface - The set of HTTP endpoints provided by the backend for communication between the client and server.
LLM	Large Language Model - An AI model (like GPT-4 or Gemini) that understands and generates text; used here to interpret user prompts and generate data extraction logic.
Regex	Regular Expression - A sequence of characters defining a search pattern. Used to extract structured data from HTML content.
Semantic Content	Structured page text with role markers (e.g. [LINK:], ## headings) produced by the headless browser for cleaner LLM input.
Celery	An asynchronous task queue/job queue based on distributed message passing. Used to process scraping and AI tasks.
Redis	In-memory data store used as a message broker for Celery and as a cache for rate-limiting.
JWT	JSON Web Token - A compact, URL-safe means of representing claims to be transferred between two parties. Used here for stateless API authentication.
Pydantic	A Python library for data validation using type annotations. FastAPI uses it to define request/response schemas (models).
Swagger UI	Automatically generated web interface for interacting with the API; provided by FastAPI at /docs.
Rate Limiting	Technique to limit the number of requests a user or IP can make in a time period, to prevent abuse. Implemented via slowapi in the API.
Playwright	A headless browser automation library used to fetch and render web pages (even with JavaScript).
Docker Compose	A tool for defining and running multi-container Docker applications. Used to orchestrate services (API, DB, Redis, etc.).
Prompt Injection	A form of attack where malicious input attempts to alter the instructions given to the LLM. The system sanitizes prompts to prevent this.
Parser Cache	A store of previously generated selectors (parsers) for reuse on repeated tasks, improving performance.

Acronyms

Acronym	Full Form	Description
API	Application Programming Interface	A set of endpoints for programmatic interaction with the backend.
UI	User Interface	The part of the system the user interacts with (none provided in this project).
CSV	Comma-Separated Values	A common text format for tabular data (can be generated from JSON results).
JSON	JavaScript Object Notation	A lightweight data-interchange format. Used for all API requests/responses.
ORM	Object-Relational Mapping	Technique for mapping database tables to code classes (SQLAlchemy is the ORM here).

Domain-Specific Terms

Web Scraping

- **Scraping Task:** A user-request to extract data from a URL based on a prompt. Represented by a ScrapingTask record.
- **Extraction Pattern:** The regex or selectors generated by the LLM to find the requested data on the page.

Data Extraction

- **Single-field Extraction:** The pipeline used when the prompt asks for one field; involves finding example text and generating a focused regex.
- **Schema Extraction:** The pipeline for multi-field requests; LLM returns structured output for all fields at once with relationships preserved.

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